

Peptimancer Vault Report

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Base Molecule Analysis

Sequence: HAEGTFTSDVSSYLEG

Length: 16 amino acids

Generation ID: d2333b08-f77a-4d74-b1bb-4030fd23cb4f

Analogue 1: Lipidated Voyager

Vault ID: PMNC-05F5151

Modified Sequence:

HAEGTFTSDVSSYLE[D-Phe16][C24]

Modifications Applied:

- D-isomer substitution at position 16 (D-Phe)
- Lipidation using a palmitic acid (C24) at position 16

IP Risk Profile:

Patent Risk: Low

Novelty Score: 80.0%

Notes: Despite sharing similarity with the existing peptide, the proposed D-isomer substitution and lipidation at position 16 significantly differentiate this analogue and lower IP risk.

Bioactivity Profile:

Binding Affinity: -9.4 kcal/mol

Predicted Half-Life: 3.0 days

Synthesis Complexity: 3/5

Estimated Cost: \$130.0 CAD/mg

Notes: D-isomer substitution at position 16 potentially enhances resistance to degradation by DPP-4. Palmitic acid can enhance the binding with albumin to improve circulatory half-life.

This report contains proprietary peptide designs. Handle confidentially.

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